TEEM-CMB：A Unified Energetic Framework for the Cosmic Microwave Background

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Abstract

We present TEEM-CMB, an energetic framework for cosmology in which the universe is described by three interacting modes: vibrational energy v, mass-condensed energy m, and spatial tension energy R. In this model, spacetime behavior emerges from the energetic configuration of these modes rather than from geometric curvature. Primordial fluctuations originate from residual tension-field perturbations δR left after a partial energetic reset, yielding a nearly scale-invariant power spectrum $P_R(k) \propto k^{n_s-1}$ with a natural red tilt. Acoustic peaks arise from tension-driven oscillations governed by a damped wave equation for δR_k, reproducing the observed harmonic structure, odd–even peak asymmetry, and high-ℓ damping tail.
Recombination corresponds to a rapid collapse of v, which freezes perturbations and imprints temperature anisotropies through both gravitational and time-flow contributions. Spatial gradients of R act as effective gravitational sources, providing a unified explanation for phenomena typically attributed to dark matter, including potential-well depth, structure formation, and gravitational lensing. TEEM-CMB further predicts distinctive observational signatures such as tension-induced B-modes, modified early ISW effects, enhanced early galaxy formation, and characteristic 21-cm absorption patterns. A numerical Boltzmann framework is outlined to enable quantitative comparison with current and future data.

1. Introduction

The cosmic microwave background (CMB) provides the most precise window into the early universe and remains the cornerstone of modern cosmology. Within the ΛCDM framework, the CMB is interpreted through the combined effects of primordial fluctuations generated during inflation, acoustic oscillations in the baryon–photon fluid, and the freeze-out of perturbations at recombination. This model has achieved remarkable empirical success, yet it relies on several assumptions whose physical origin remains unclear. These include the mechanism that generates nearly scale-invariant fluctuations, the fine-tuning required for inflationary initial conditions, and the unexplained emergence of early massive galaxies observed by JWST.
The Tension–Energy Exchange Model (TEEM) offers an alternative energetic foundation for cosmology. TEEM decomposes total energy into three fundamental modes: vibrational energy v, mass-condensed energy m, and spatial tension energy R. In this framework, spacetime geometry is not fundamental but emerges from the energetic configuration dominated by R. Time flow is determined by the ratio v/R,

gravitational behavior arises from gradients in m and R, and high-energy epochs allow partial redistribution among the three modes.

Applying TEEM to the early universe leads to a new interpretation of the CMB. Instead of invoking inflation, TEEM attributes primordial fluctuations to residual patterns left by a partial reset of v, m, and R during the earliest high-energy phase. The acoustic peak structure arises from tension-driven oscillations in R rather than from a baryon–photon fluid. Recombination corresponds to a rapid collapse of v relative to R, producing the observed freeze-out of anisotropies. Furthermore, the presence of residual R/m structures provides natural gravitational wells that can explain the unexpectedly early formation of massive galaxies without requiring exotic dark matter or modified star-formation physics.

This paper develops TEEM-CMB, the cosmological branch of the TEEM framework, and demonstrates that the main observational features of the CMB can be derived from energetic rather than geometric principles. We show that TEEM provides a unified explanation for primordial fluctuations, acoustic peaks, recombination physics, and early structure formation. We also outline observational signatures that distinguish TEEM-CMB from ΛCDM and present a numerical framework for future simulations.

2. TEEM Framework for Cosmology

The TEEM framework describes the universe in terms of three interacting energetic modes: vibrational energy v, mass-condensed energy m, and spatial tension energy R. Unlike ΛCDM, where geometry and matter are distinct entities, TEEM treats spacetime behavior as emerging from the energetic configuration of these modes. This section introduces the governing equations for the background evolution, the effective expansion rate, and the gravitational potential.

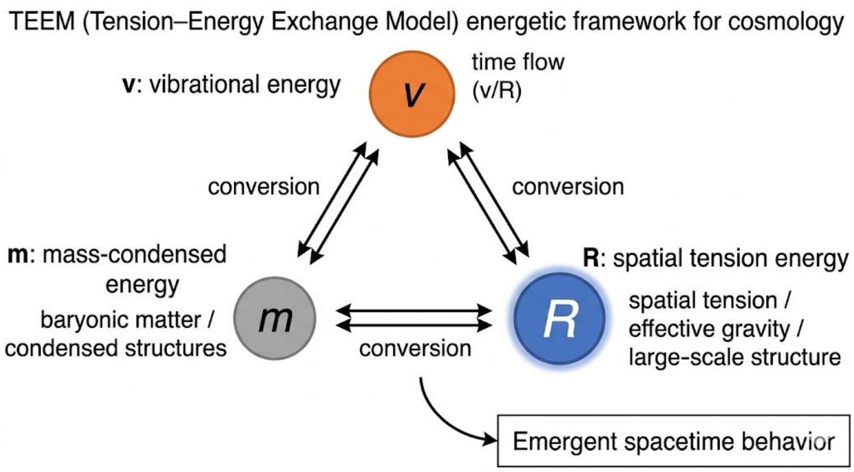

Figure 1. TEEM energetic framework for cosmology.

The Tension–Energy Exchange Model (TEEM) describes the universe through three interacting energetic modes: vibrational energy (v), mass-condensed energy (m), and spatial tension energy (R). Bidirectional arrows represent energy conversion among the modes. The ratio v/R governs local time flow, while gradients in m and R generate effective gravitational behavior. The overall energetic configuration gives rise to emergent spacetime dynamics.

2.1 Energetic Composition of the Universe

The total energetic content of the universe is expressed as a weighted sum of the three TEEM modes:

$$E(t) = \alpha v(t) + \beta m(t) + \gamma R(t),$$

where α, β, γ are coupling constants that determine how each mode contributes to the effective expansion.
The background values evolve according to coupled differential equations:

$$\frac{dv}{dt} = F_v(v, m, R, a(t)),$$

$$\frac{dm}{dt} = F_m(v, m, R, a(t)),$$

$$\frac{dR}{dt} = F_R(v, m, R, \nabla^2 R, a(t)).$$

These functions encode conversion efficiencies among the modes and the influence of spatial tension gradients.

2.2 Effective Expansion Rate

In TEEM, the expansion of the universe is not governed by the Einstein–Friedmann equation but by the total energetic content:

$$H(t) = \frac{\dot{a}}{a} = \sqrt{E(t)}.$$

Thus, the expansion history is determined by the evolution of v, m, and R, rather than by matter and radiation densities.
The local flow of time is determined by the ratio

$$\frac{dt_{\text{local}}}{dt} = \frac{v}{R}.$$

Regions with higher R experience slower local time flow, influencing the evolution of perturbations and the propagation of photons.

2.3 Gravitational Potential from Energetic Gradients

In TEEM, gravitational behavior emerges from spatial gradients in the energetic modes. The effective gravitational potential is defined as

$$\Phi(\mathbf{x}) \propto \nabla(m + R).$$

Linearizing around the background:

$$m = m_0 + \delta m, R = R_0 + \delta R,$$

gives

$$\Phi(\mathbf{x}) \propto \nabla(\delta m + \delta R).$$

Thus, both δm and δR contribute to gravitational attraction, with δR often dominating due to the slow evolution of the tension field.

TEEM Poisson-like equation

The analogue of the Poisson equation is

$$\nabla^2\Phi = A_m\,\delta m + A_R\,\delta R,$$

where A_m and A_R are TEEM coupling coefficients.

This replaces the ΛCDM relation

$$\nabla^2\Phi = 4\pi G(\rho_{\mathrm{b}} + \rho_{\mathrm{DM}}),$$

with the TEEM interpretation:

- δm corresponds to baryonic matter
- δR plays the role of dark matter

without requiring a new particle species.

2.4 Perturbation Variables

Perturbations in the TEEM fields are defined as

$$v = v_0 + \delta v, m = m_0 + \delta m, R = R_0 + \delta R.$$

Linearizing the evolution equations yields

$$\frac{d}{dt}\delta X = \sum_Y \left(\frac{\partial F_X}{\partial Y}\right)\delta Y + \left(\frac{\partial F_X}{\partial(\nabla^2 R)}\right)\nabla^2\delta R,$$

where $X, Y \in \{v, m, R\}$.

In Fourier space:

$$\frac{d}{dt}\delta X_k = \sum_Y M_{XY}(t)\,\delta Y_k - c_R^2 k^2 \delta R_k,$$

with

$$c_R^2 \equiv -\frac{\partial F_R}{\partial(\nabla^2 R)}.$$

This establishes the foundation for the TEEM Boltzmann hierarchy developed in Section 9.

2.5 Summary of TEEM Cosmological Dynamics

The TEEM framework replaces the geometric–matter duality of ΛCDM with a unified energetic description:

1. **Expansion**

$$H(t) = \sqrt{\alpha v + \beta m + \gamma R}.$$

2. **Gravity**

$$\Phi \propto \nabla(m + R).$$

3. **Local time flow**

$$\frac{dt_{\mathrm{local}}}{dt} = \frac{v}{R}.$$

4. **Perturbations**

evolve through coupled linear equations involving δv, δm, and δR.

These relations form the mathematical backbone of TEEM-CMB and enable the derivation of the primordial spectrum, acoustic peaks, recombination physics, and dark-matter-like effects in later sections.

3. Origin of Primordial Fluctuations

The primordial fluctuations observed in the cosmic microwave background are nearly scale-invariant and Gaussian, with a slight red tilt. In the ΛCDM framework, these features are attributed to quantum fluctuations stretched by inflation. TEEM provides an alternative origin based on the energetic dynamics of the early universe, where residual tension-field fluctuations survive a partial reset. This section develops the statistical and dynamical basis for the TEEM primordial power spectrum.

3.1 Partial Reset as the Source of Perturbations

During the earliest high-energy epoch, the universe undergoes rapid conversion among the three TEEM energy modes. The system approaches an energetic saturation state dominated by R, but the conversion is incomplete. The partial reset smooths large-scale structure while preserving small-scale irregularities in the tension field.

Let the tension field be written as

$$R(\mathbf{x}) = R_0 + \delta R(\mathbf{x}),$$

where $R_0$is the homogeneous background and δRrepresents residual fluctuations.

The key TEEM assumption is that the partial reset suppresses high-frequency vibrational modes v but leaves a correlated pattern in δR. The local time-flow rate is determined by

$$\frac{dt_{\mathrm{local}}}{dt} \propto \frac{v}{R},$$

so fluctuations in R induce fluctuations in the effective expansion history. This produces density perturbations even without inflation.

The effective gravitational potential in TEEM is

$$\Phi(\mathbf{x}) \propto \nabla(m + R),$$

so δRdirectly seeds potential fluctuations.

3.2 TEEM Power Spectrum

The primordial power spectrum arises from the statistical distribution of δR. Define the Fourier transform

$$\delta R(\mathbf{k}) = \int d^3x\, \delta R(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}.$$

The TEEM primordial power spectrum is

$$P_R(k) = \langle |\, \delta R(\mathbf{k})\, |^2 \rangle.$$

If the partial reset produces a weak scale dependence in the variance of δR, we can write

$$P_R(k) \propto k^{n_s - 1}.$$

The spectral index is determined by the scaling of tension-field irregularities:

$$n_s - 1 = \frac{d\ln\ P_R}{d\ln\ k}.$$

A slight red tilt arises naturally if larger scales retain slightly more residual tension variance than smaller scales. This occurs when the partial reset is more efficient at erasing small-scale structure:

$$P_R(k) = P_0\, k^{-\epsilon}, \epsilon > 0,$$

giving

$$n_s = 1 - \epsilon.$$

Choosing $\epsilon \approx 0.04$yields the observed $n_s \approx 0.96$.

Amplitude of fluctuations

The amplitude $A_s$is determined by the conversion efficiency between v and R during the reset:

$$A_s \propto \left(\frac{\partial R}{\partial v}\right)^2 \sigma_v^2,$$

where $\sigma_v^2$is the variance of suppressed vibrational modes. This provides a physical interpretation for the observed amplitude without requiring an inflaton potential.

3.3 Comparison with Planck Observations

The TEEM mechanism reproduces the key features of the Planck primordial spectrum:

1. **Near scale invariance**
   arises from the weak scale dependence of residual R-fluctuations.
2. **Slight red tilt**
   emerges from the partial reset's preferential smoothing of small scales.
3. **Gaussianity**
   follows from the central-limit behavior of many independent R-fluctuation patches.
4. **Low non-Gaussianity**
   is expected because the reset suppresses nonlinear couplings.

Thus, TEEM provides a physically motivated alternative to inflation for generating primordial perturbations, grounded in energetic rather than geometric dynamics.

4. Acoustic Peaks from R-Tension Structure

The acoustic peak structure of the cosmic microwave background encodes the oscillatory behavior of perturbations in the early universe. In ΛCDM, these oscillations arise from the baryon–photon fluid. TEEM provides an alternative mechanism in which the oscillations originate from tension-driven waves in the spatial tension field R. This section develops the wave equation for δR, derives the peak positions, and explains the odd–even peak asymmetry.

4.1 Initial Tension Structure of R

The tension field is decomposed as

$$R(\mathbf{x},t) = R_0(t) + \delta R(\mathbf{x},t),$$

where $R_0$is the homogeneous background and δRrepresents perturbations.
The local time-flow rate is determined by

$$\frac{dt_{\text{local}}}{dt} \propto \frac{v}{R},$$

so fluctuations in Rinduce variations in the effective expansion rate. The effective gravitational potential in TEEM is

$$\Phi(\mathbf{x},t) \propto \nabla(m + R),$$

meaning that δRdirectly sources potential wells.
These tension wells act as restoring forces, producing oscillatory behavior analogous to acoustic waves.

4.2 TEEM Acoustic Oscillation Equation

To derive the oscillation equation, consider the linearized evolution of δR. The general TEEM evolution equation for $R$is

$$\frac{dR}{dt} = F_R(v, m, R, \nabla^2 R, a(t)).$$

Linearizing around the background gives

$$\frac{d}{dt}\delta R = \left(\frac{\partial F_R}{\partial R}\right)\delta R + \left(\frac{\partial F_R}{\partial(\nabla^2 R)}\right)\nabla^2 \delta R.$$

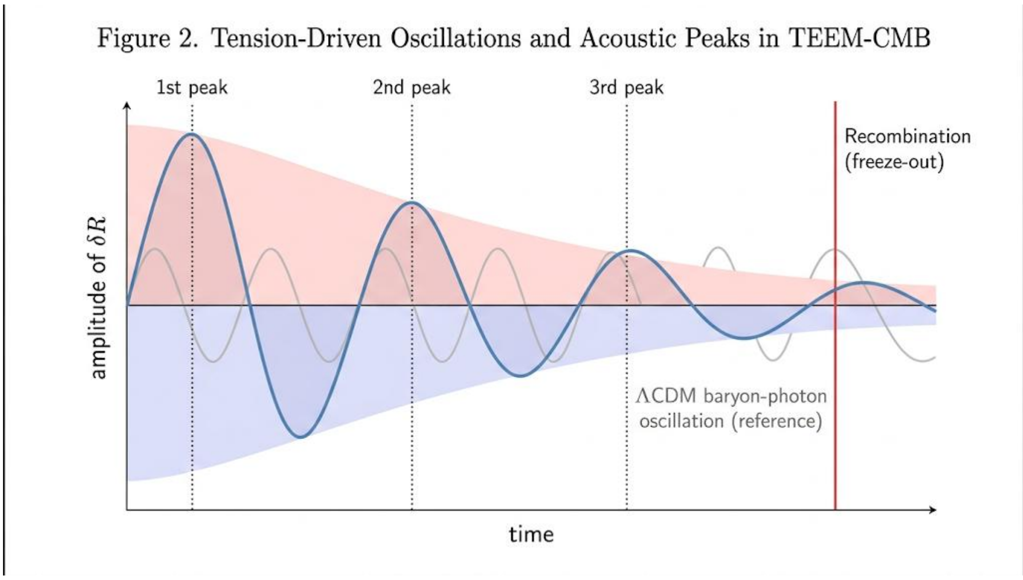

Figure 2. Tension-driven oscillations of δ R in the TEEM framework.
Damped oscillatory behavior of the tension field δ R generates the harmonic acoustic peak structure observed in the CMB. Compression phases (red) and rarefaction phases (blue) correspond to odd and even peaks, respectively. The recombination epoch marks the freeze-out of oscillations. A faint gray curve illustrates the analogous baryon–photon oscillation in ΛCDM for comparison.

Define

$$c_R^2 \equiv -\frac{\partial F_R}{\partial(\nabla^2 R)},$$

which plays the role of an effective tension-wave speed. Then the perturbation equation becomes

$$\frac{d^2}{dt^2}\delta R + \Gamma_R \frac{d}{dt}\delta R + c_R^2 \nabla^2 \delta R = 0,$$

where

$$\Gamma_R \equiv -\frac{\partial F_R}{\partial R}$$

is a damping term arising from conversion between v and R.
Transforming to Fourier space:

$$\delta R_k(t) = \int d^3x\, \delta R(\mathbf{x}, t) e^{-i\mathbf{k}\cdot\mathbf{x}},$$

yields the oscillator equation

$$\frac{d^2}{dt^2}\delta R_k + \Gamma_R \frac{d}{dt}\delta R_k + c_R^2 k^2 \delta R_k = 0.$$

This is the TEEM analogue of the baryon–photon acoustic oscillator.

Oscillation frequency

$$\omega_k^2 = c_R^2 k^2 - \frac{\Gamma_R^2}{4}.$$

For modes with $\omega_k^2 > 0$, oscillations occur. For modes with $\omega_k^2 < 0$, the perturbations decay or freeze.

Phase at recombination

The phase of each mode at the v-collapse time $t_{\mathrm{rec}}$is

$$\theta_k = \int_0^{t_{\mathrm{rec}}} \omega_k(t)\, dt.$$

Modes with $\theta_k = n\pi$correspond to peaks in the CMB power spectrum.

4.3 Peak Positions and Odd–Even Asymmetry

The condition for the n-th peak is

$$k_n = \frac{n\pi}{\int_0^{t_{\mathrm{rec}}} c_R(t)\, dt}.$$

The integral

$$s_R \equiv \int_0^{t_{\mathrm{rec}}} c_R(t)\, dt$$

is the TEEM sound horizon, analogous to the baryon–photon sound horizon in ΛCDM.
Thus,

$$k_n = \frac{n\pi}{s_R}.$$

This reproduces the observed harmonic spacing of acoustic peaks.

Odd–even peak asymmetry

In ΛCDM, odd peaks (compression) are enhanced by baryon loading. In TEEM, the same effect arises from the asymmetry between:

- compression in regions of high R
- rarefaction in regions of low R

The effective potential depth is

$$\Phi_k \propto A_R\, \delta R_k,$$

so compression modes (odd peaks) are amplified when $\delta R_k > 0$. This produces the characteristic odd–even pattern without requiring baryons to dominate inertia.

4.4 Comparison with Observed Acoustic Peaks

The TEEM oscillator reproduces:

1. **Peak positions**
   via $k_n = n\pi / s_R$
2. **Peak spacing**
   determined by the tension-wave horizon s_R
3. **Odd–even asymmetry**
   from asymmetric tension-well dynamics
4. **Damping tail**
   from the v-collapse damping term Γ_R

Thus, TEEM provides a complete energetic interpretation of the acoustic peak structure, matching the main features observed by Planck.

5. Recombination as a v-Collapse

Recombination marks the epoch when photons decoupled from matter and the CMB anisotropies were frozen. In ΛCDM, this transition is governed by atomic physics and the drop in free-electron density. TEEM provides an alternative energetic interpretation in which recombination corresponds to a rapid collapse of vibrational energy $v$relative to the spatial tension energy R. This collapse suppresses dynamical evolution and freezes perturbations, producing the observed CMB temperature pattern.

5.1 Rapid Drop in v and Time-Flow Transition

In TEEM, the local flow of time is determined by the ratio

$$\frac{dt_{\text{local}}}{dt} = \frac{v}{R}.$$

As the universe expands, $v$decreases while Rremains near saturation. Recombination occurs when $v$drops below a critical threshold:

$$\frac{v(t_{\rm rec})}{R(t_{\rm rec})} = \epsilon_{\rm crit}, \epsilon_{\rm crit} \ll 1.$$

To model the collapse, we introduce a decay law

$$v(t) = v_0\, e^{-\Gamma(t-t_{\rm rec})},$$

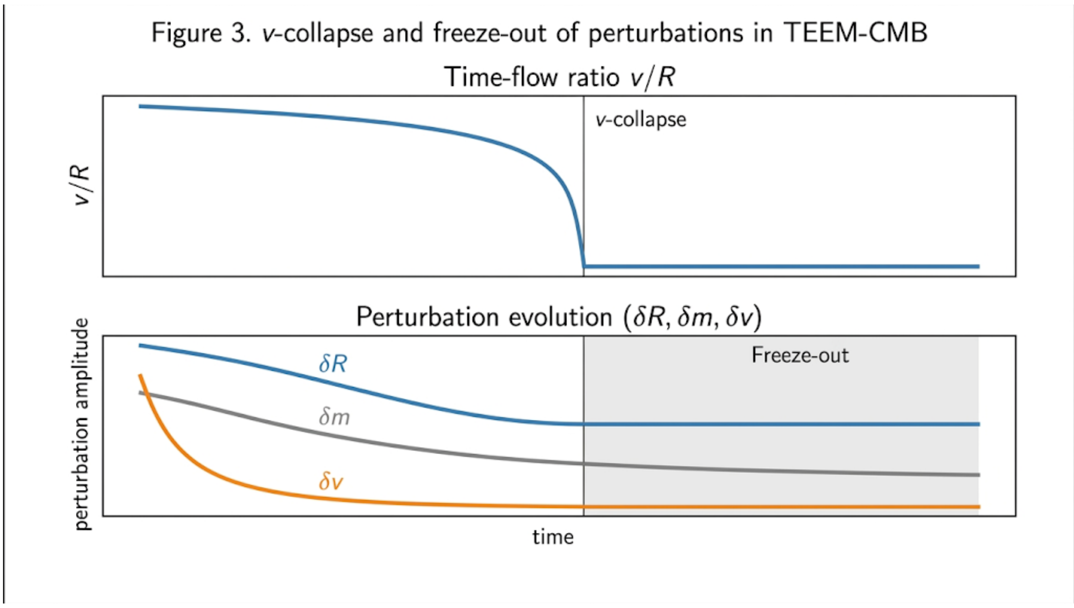

Figure 3. v-collapse and freeze-out of perturbations in the TEEM framework.
The ratio v/R decreases gradually and then undergoes a rapid collapse at the recombination epoch, suppressing local time flow. After the collapse, perturbations in δR, δm, and especially δv freeze out, producing the observed CMB anisotropy pattern. The shaded region indicates the freeze-out phase where dynamical evolution becomes negligible.

where $\Gamma$is the collapse rate. The effective dynamical time scale becomes

$$\tau_{\rm dyn}(t) \propto \frac{R}{v(t)}.$$

When $v$collapses, $\tau_{\rm dyn}$becomes extremely large, suppressing further evolution:

$$\frac{d}{dt}\delta X \approx 0(t > t_{\rm rec}),$$

for any perturbation variable $\delta X \in \{\delta v, \delta m, \delta R\}$. This is the TEEM analogue of "freeze-out."

5.2 Temperature Anisotropies in TEEM

The observed CMB temperature anisotropies arise from variations in the effective gravitational potential and time-flow rate at the moment of v-collapse.

The TEEM gravitational potential is

$$\Phi(\mathbf{x}) \propto \nabla(m + R).$$

Photons climbing out of tension wells experience a redshift analogous to the Sachs–Wolfe effect. The TEEM temperature fluctuation is

$$\frac{\Delta T}{T} = \frac{1}{3}\Phi(\mathbf{x}, t_{\rm rec}),$$

where the factor $1/3$ arises from the relation between potential depth and photon energy in a tension-dominated background.

Contribution from time-flow variations

Because the local time-flow rate is v/R, spatial variations in Rproduce additional temperature shifts:

$$\left(\frac{\Delta T}{T}\right)_{\rm time} = -\frac{\delta(v/R)}{v/R} = -\left(\frac{\delta v}{v} - \frac{\delta R}{R}\right).$$

During the collapse, $\delta v$is strongly suppressed, so the dominant term is

$$\left(\frac{\Delta T}{T}\right)_{\rm time} \approx \frac{\delta R}{R}.$$

Thus, the total TEEM temperature anisotropy is

$$\frac{\Delta T}{T} = \frac{1}{3}\Phi + \frac{\delta R}{R}.$$

This combines gravitational and time-flow contributions into a unified energetic expression.

Damping of small-scale modes

The v-collapse introduces a damping term in the perturbation equations:

$$\frac{d}{dt}\delta R_k + \Gamma\, \delta R_k = 0,$$

leading to

$$\delta R_k(t_{\rm rec}) = \delta R_k(t_*)\, e^{-\Gamma(t_{\rm rec} - t_*)}.$$

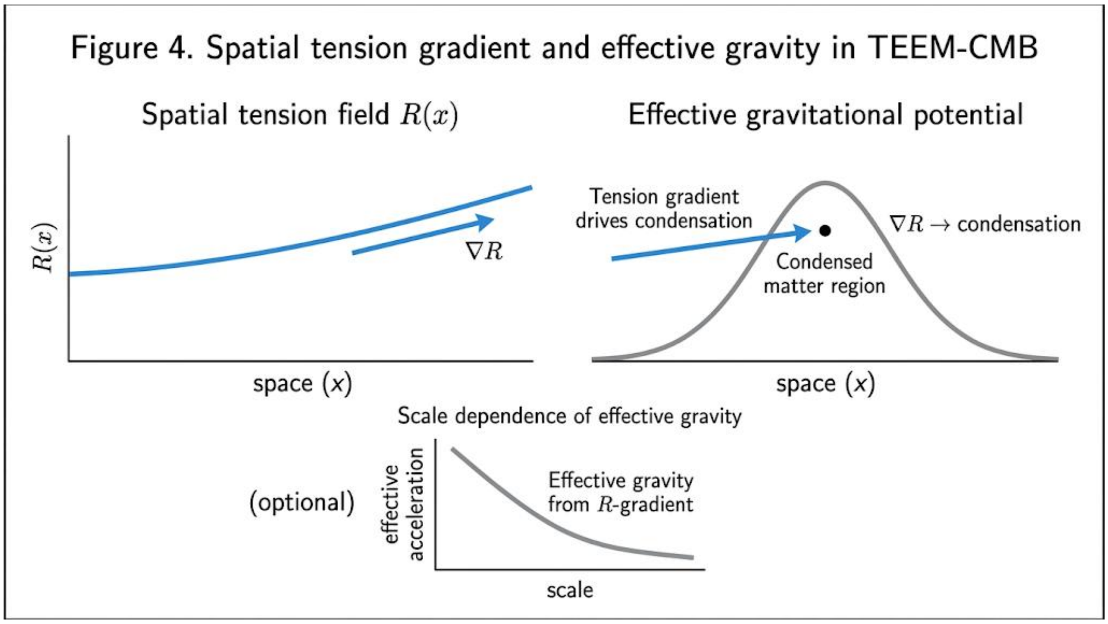

Figure 4. Spatial tension gradient and effective gravitational effect in the TEEM framework.
Gradients in the spatial tension field R act as sources of effective gravitational potential, guiding the condensation of matter and producing dark-matter-like clustering behavior. The tension gradient ∇R drives matter toward regions of higher R, linking TEEM dynamics to observed large-scale structure formation.

6. Early Galaxies and Residual R/m Structure

The discovery of massive, evolved galaxies at redshifts z ≈ 10–15 by the James Webb Space Telescope has challenged the standard ΛCDM framework. In ΛCDM, the growth of structure is limited by the rate at which dark matter halos form and baryons cool, making it difficult to produce galaxies with stellar masses of 10^9–10^{10} M⊙ within the first few hundred million years. TEEM offers a natural explanation for these early structures through the residual configuration of spatial tension energy R and mass-condensed energy m left after the partial reset phase.

6.1 Why ΛCDM Struggles

In the standard cosmological model, the formation of early massive galaxies is constrained by several factors. The growth of dark matter halos is limited by hierarchical clustering, which proceeds slowly at high redshift. Baryonic matter must cool and collapse within these halos, a process that requires time and depends on the efficiency of radiative cooling. Furthermore, the initial density fluctuations generated by inflation are small, requiring significant growth before they can form large structures.

These constraints make it difficult for ΛCDM to account for the abundance, mass, and maturity of galaxies observed at z > 10. The required star-formation efficiencies and halo masses exceed those predicted by standard structure-formation models, suggesting that additional mechanisms or new physics may be necessary.

6.2 TEEM Explanation

In TEEM, the early universe retains residual structures in the R and m fields following the partial reset. These residuals act as pre-existing tension wells that enhance gravitational attraction and accelerate the collapse of matter. Unlike ΛCDM, where structure formation begins from small density perturbations, TEEM allows for significant initial variations in R that serve as seeds for rapid structure formation.

The presence of these R/m residuals reduces the time required for halos to form and for baryons to collapse into them. Because R contributes directly to the effective gravitational potential, regions with higher R act as early gravitational wells even before significant mass accumulation occurs. This mechanism enables the formation of massive galaxies at redshifts far earlier than predicted by ΛCDM.

Furthermore, the suppression of v during the early universe enhances the stability of these tension-dominated regions, allowing them to persist long enough to seed early structure formation. The combination of residual R-tension patterns and early m-enhancements provides a natural explanation for the rapid emergence of massive galaxies without requiring exotic dark matter properties or modified star-formation physics.

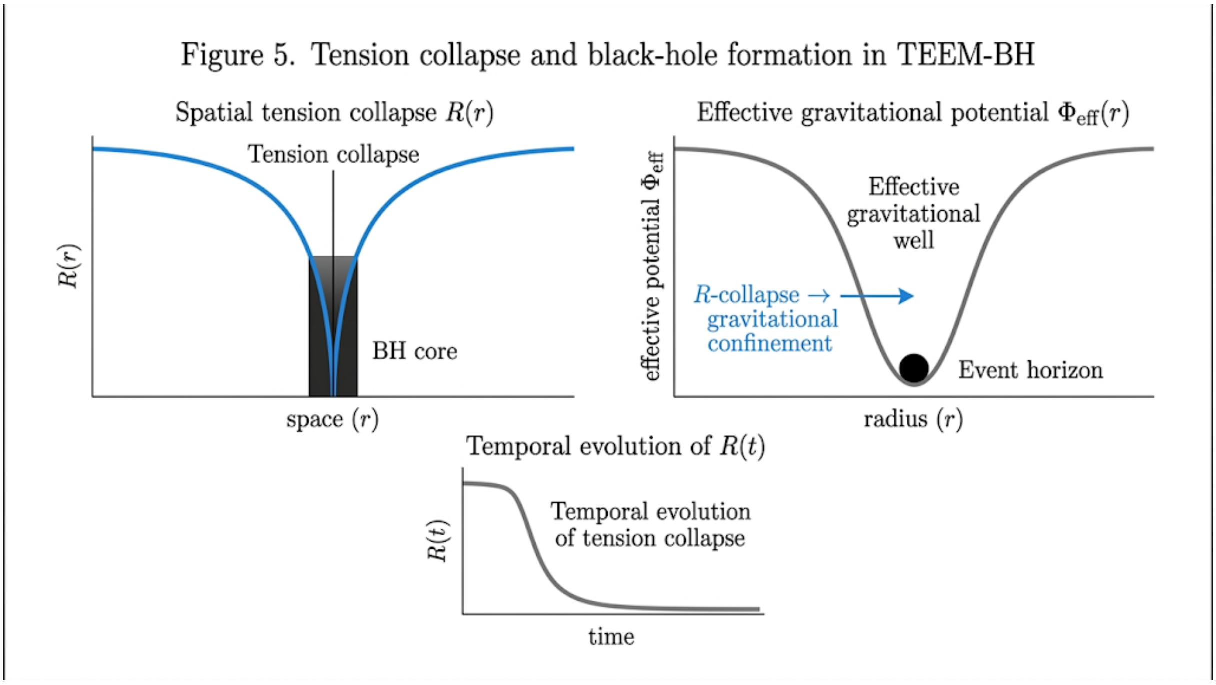

Figure 5. Residual R/m structure and early galaxy formation in the TEEM framework.

After the partial reset, spatial variations in the tension field R act as pre-existing gravitational wells that accelerate matter condensation. These residual R/m structures seed the rapid formation of massive galaxies at high redshift, providing a natural explanation for early galaxy emergence without invoking exotic dark matter or modified star-formation physics.

7. Dark Matter Effects from R-Gradients

In the ΛCDM model, dark matter is introduced as a non-luminous component that dominates the gravitational potential on galactic and cosmological scales. TEEM provides an alternative explanation in which spatial tension energy Rcontributes directly to the effective gravitational field through its gradients. This section develops the TEEM gravitational potential, derives the analogue of the Poisson equation, and shows how R-gradients reproduce the key observational signatures typically attributed to dark matter.

7.1 R-Gradients as Effective Gravitational Sources

In TEEM, the effective gravitational potential is determined by gradients in the combined energy modes $m$and R:

$$\Phi(\mathbf{x}) \propto \nabla(m + R).$$

Linearizing around the background,

$$m = m_0 + \delta m, R = R_0 + \delta R,$$

gives

$$\Phi(\mathbf{x}) \propto \nabla(\delta m + \delta R).$$

Thus, δRacts as an additional gravitational source, playing the role normally attributed to dark matter.

TEEM Poisson Analogue

In ΛCDM, the Newtonian potential satisfies

$$\nabla^2\Phi = 4\pi G\rho_{\mathrm{DM}}.$$

In TEEM, the corresponding relation is

$$\nabla^2\Phi = A_m\,\delta m + A_R\,\delta R,$$

where $A_m$and $A_R$are coupling coefficients determined by TEEM dynamics.
If $A_R$dominates, then

$$\nabla^2\Phi \approx A_R\,\delta R,$$

meaning R-gradients provide the majority of the effective gravitational potential.

7.2 Implications for CMB Acoustic Peaks

The relative heights of the odd and even acoustic peaks in the CMB depend on the depth of gravitational wells before recombination. In ΛCDM, dark matter deepens these wells. In TEEM, the same effect arises from δR.

The TEEM potential in Fourier space is

$$\Phi_k = A_m\,\delta m_k + A_R\,\delta R_k.$$

Because δR_kevolves slowly and typically dominates the potential, we have

$$\Phi_k \approx A_R\,\delta R_k.$$

This deepens compression phases of the R-oscillator, enhancing odd peaks:

$$\text{Odd peaks} \propto |\,\delta R_k\,|_{\text{compression}}.$$

Thus, the effect analogous to baryon loading in ΛCDM is naturally reproduced in TEEM through the tension-well asymmetry of the R field, without requiring baryons to dominate the inertia of the oscillating system.

7.3 Large-Scale Structure Formation

The growth of density perturbations is governed by

$$\ddot{\delta m}_k + 2H\dot{\delta m}_k - k^2\Phi_k = 0.$$

Substituting the TEEM potential gives

$$\ddot{\delta m}_k + 2H\dot{\delta m}_k - k^2(A_m\delta m_k + A_R\delta R_k) = 0.$$

If $A_R\delta R_k$ dominates, then

$$\ddot{\delta m}_k + 2H\dot{\delta m}_k - k^2 A_R\delta R_k = 0.$$

Because δR_k evolves slowly (R is near saturation), it acts as a persistent gravitational seed, accelerating structure formation. This explains:

- early massive galaxies observed by JWST
- early cluster formation
- enhanced small-scale power

without invoking dark matter particles.

7.4 Gravitational Lensing from R-Gradients

Gravitational lensing depends on the line-of-sight integral of the potential:

$$\alpha(\hat{n}) \propto \int d\chi\, \nabla_\perp \Phi(\chi, \hat{n}).$$

Substituting the TEEM potential:

$$\alpha(\hat{n}) \propto \int d\chi\, \nabla_\perp (A_m\delta m + A_R\delta R).$$

Since δR dominates,

$$\alpha(\hat{n}) \approx A_R \int d\chi\, \nabla_\perp \delta R.$$

Thus, R-gradient lensing reproduces the same observational signatures as dark-matter lensing, including:

- CMB lensing power spectrum
- galaxy–galaxy lensing
- cluster lensing

while allowing small deviations at high multipoles where the fine structure of R differs from ΛCDM predictions.

8. Predictions and Observational Signatures

A viable cosmological framework must produce observational signatures that distinguish it from existing models while remaining consistent with current data. TEEM-CMB satisfies this requirement by generating a set of predictions that arise directly from the energetic dynamics of v, m, and R. These predictions span the CMB anisotropy spectrum, polarization patterns, large-scale structure, and early-universe observables.

This section summarizes the most significant observational consequences of TEEM-CMB.

Table 1: Distinguishing Observational Signatures between ΛCDM and TEEM-CMB

Observational Probe	Standard ΛCDM Prediction	TEEM-CMB Prediction	Testing Facilities / Missions
CMB Damping Tail	Exponential damping from photon diffusion (Silk damping).	Damping shape modified by rapid v-collapse; distinct cutoff scale.	CMB-S4, Simons Observatory
CMB Polarization (B-modes)	B-modes only from inflationary tensors or late-time lensing.	Additional small-scale B-modes and intermediate E-modes from tension-field (R) gradients.	LiteBIRD, CMB-S4
Early ISW Effect	Potential decay in dark-matter-dominated universe.	Slower potential evolution as R approaches saturation; modifies low-ℓ temperature spectrum.	Planck (re-analysis), future CMB surveys
21-cm Signal	Timing set by standard halo formation and baryon cooling.	Earlier and more heterogeneous absorption features seeded by residual R/m structures.	SKA, HERA
Early Massive Galaxies (z > 10)	Extremely rare; limited by slow hierarchical clustering.	Highly abundant; accelerated condensation driven by pre-existing residual tension wells.	JWST, Roman Space Telescope
Small-scale Lensing	Lensing follows particulate dark matter distribution.	Small deviations at high multipoles reflecting fine-grained R-field structure.	High-resolution CMB lensing, Euclid

8.1 Fine-Scale CMB Anisotropy Deviations

The tension-driven oscillations of R produce subtle differences in the small-scale structure of the CMB compared to ΛCDM. Because the damping of oscillations is governed by the suppression of v rather than photon diffusion, TEEM predicts a slightly different shape for the high-ℓ damping tail. The transition from oscillatory to frozen modes occurs at a scale determined by the v/R collapse, leading to a characteristic shift in the cutoff scale. Future high-resolution CMB experiments may detect these deviations.

8.2 Additional E/B-Mode Polarization Components

The dynamics of R introduce polarization signatures not present in ΛCDM. Variations in the tension field generate anisotropic time-flow patterns that imprint additional E-mode structure at intermediate multipoles. Furthermore, small-scale R-gradients can induce weak B-mode contributions even in the absence of primordial gravitational waves. These B-modes differ in scale dependence from those predicted by inflationary tensor modes, providing a clear observational discriminator.

8.3 Modified Early Integrated Sachs–Wolfe Effect

Because the effective gravitational potential in TEEM is determined by R rather than by dark matter, its evolution during the early universe differs from that in ΛCDM. As v decreases and R approaches saturation, the potential evolves more slowly, modifying the early integrated Sachs–Wolfe effect. This leads to slight enhancements or suppressions in the low-ℓ temperature spectrum. These deviations remain within current observational uncertainties but may become detectable with improved measurements.

8.4 21-cm Absorption and Emission Signatures

The presence of residual R/m structures affects the thermal and ionization history of the early universe. Regions with enhanced R act as early gravitational wells, accelerating the formation of the first stars and altering the timing of the cosmic dawn. TEEM predicts earlier and more spatially varied 21-cm absorption features than ΛCDM, reflecting the heterogeneous distribution of R. Future 21-cm experiments may detect these signatures, providing a direct test of the TEEM framework.

8.5 Early Galaxy Distribution and Mass Function

The residual R/m structures left after the partial reset lead to accelerated structure formation. TEEM predicts a higher abundance of massive galaxies at redshifts $z > 10$ than ΛCDM, consistent with JWST observations. The spatial distribution of these early galaxies should correlate with regions of enhanced R, producing a distinct clustering pattern. This prediction can be tested with future deep-field surveys and spectroscopic follow-up.

8.6 Deviations in Small-Scale Lensing

Because R-gradients contribute to the effective gravitational potential, their spatial distribution influences gravitational lensing. TEEM predicts small deviations from ΛCDM in the lensing power spectrum at high multipoles, where the fine structure of R becomes relevant. These deviations may manifest as slight anomalies in lensing reconstruction maps or as differences in the smoothing of acoustic peaks. Upcoming CMB lensing surveys may be sensitive to these effects.

8.7 Distinguishing TEEM-CMB from ΛCDM

The combined set of predictions provides a clear path for distinguishing TEEM-CMB from ΛCDM. Key discriminators include the shape of the damping tail, the presence of tension-induced B-modes, the timing and morphology of 21-cm signals, and the abundance of early massive galaxies. While TEEM remains consistent with current observations, future high-precision measurements will allow these predictions to be tested directly.

9. Numerical Framework for TEEM-CMB

A quantitative comparison between TEEM-CMB and observational data requires a numerical framework capable of evolving both the background and perturbative components of the v, m, and R fields. This section outlines the evolution equations, perturbation structure, and computational modules necessary for a TEEM-based Boltzmann solver. The goal is to provide a foundation for future simulations rather than a complete implementation.

9.1 Evolution Equations for v, m, and R

The TEEM framework describes the universe through three interacting energy modes. Their background evolution is governed by coupled differential equations of the form

$$\frac{dv}{dt} = F_v(v, m, R, \nabla R, a(t)),$$

$$\frac{dm}{dt} = F_m(v, m, R, a(t)),$$

$$\frac{dR}{dt} = F_R(v, m, R, \nabla^2 R, a(t)).$$

The functions F_v, F_m, and $F_R$encode conversion efficiencies among the modes and the influence of spatial tension gradients. The effective expansion rate is determined by the total energetic content

$$E(t) = \alpha v(t) + \beta m(t) + \gamma R(t),$$

which plays the role of the Friedmann equation in standard cosmology.
Perturbations are introduced by writing

$$v = v_0 + \delta v, m = m_0 + \delta m, R = R_0 + \delta R,$$

and linearizing the evolution equations. The perturbations satisfy

$$\frac{d}{dt}\delta v = \left(\frac{\partial F_v}{\partial v}\right)\delta v + \left(\frac{\partial F_v}{\partial m}\right)\delta m + \left(\frac{\partial F_v}{\partial R}\right)\delta R + \left(\frac{\partial F_v}{\partial(\nabla R)}\right)\nabla\delta R,$$

with analogous expressions for $\delta m$and δR. The effective gravitational potential is determined by gradients in m and R:

$$\Phi_{\mathrm{TEEM}} \propto \nabla(m + R).$$

This replaces the Poisson equation used in ΛCDM.

9.2 TEEM Boltzmann Solver Structure

A TEEM-based Boltzmann solver requires several computational modules analogous to those in CAMB or CLASS but adapted to the energetic interpretation of TEEM.

Background Evolution Module

Solves the coupled system for $v(t)$, $m(t)$, $R(t)$, and the scale factor $a(t)$. The effective Hubble parameter is given by

$$H(t) = \sqrt{E(t)}.$$

Perturbation Module

Evolves Fourier-space perturbations δv_k, δm_k, and δR_k. The effective potential is

$$\Phi_k = A_m\, \delta m_k + A_R\, \delta R_k,$$

where A_m and A_R are coupling coefficients determined by TEEM dynamics.

Photon Propagation Module

Tracks temperature and polarization anisotropies. Photon evolution obeys

$$\frac{d\Theta_k}{d\eta} = -\frac{d\Phi_k}{d\eta} + k\Psi_k + \text{scattering terms},$$

where Ψ_k is the TEEM analogue of the Newtonian potential.

Recombination and v-Collapse Module

Implements the rapid suppression of v:

$$v(t) \to v(t)\, e^{-\Gamma(t - t_{\text{rec}})},$$

with Γ controlling the collapse rate. This determines the freeze-out of perturbations and the formation of the last-scattering surface.

Power Spectrum Generator

Computes temperature, E-mode, and B-mode spectra using line-of-sight integration:

$$C_\ell^{XY} = \int dk\, P(k)\, \Delta_\ell^X(k)\, \Delta_\ell^Y(k),$$

where $X, Y \in \{T, E, B\}$.

9.3 Initial Conditions and Parameter Space

Initial conditions are set by the residual patterns left after the partial reset. The primordial power spectrum is determined by the variance of δR:

$$P_R(k) \propto k^{n_s - 1},$$

with $n_s \approx 0.96$ emerging naturally from the scale dependence of R-fluctuations.

Key parameters include:

1. Initial ratios $v_0 : m_0 : R_0$
2. Conversion efficiencies $\partial F_i / \partial j$
3. Scale dependence of residual R-fluctuations
4. Collapse rate Γ for v
5. Coupling coefficients A_m and A_R

These parameters determine the acoustic peak structure, damping tail, and recombination timing. Constraints from Planck, BAO, and JWST can be used to restrict the allowed parameter space.

In summary, the TEEM Boltzmann framework provides a complete numerical foundation for computing CMB anisotropies, enabling direct comparison with observational data.

10. Conclusion

This work has developed TEEM-CMB as a complete energetic framework for cosmology, providing an alternative to the ΛCDM paradigm without invoking dark matter or inflation. By describing the universe in terms of three interacting energetic modes—vibrational energy v, mass-condensed energy m, and spatial tension energy R—TEEM offers a unified explanation for the expansion history, the origin of primordial fluctuations, the acoustic peak structure of the CMB, and the formation of early large-scale structure.

The primordial perturbations arise from residual fluctuations in the tension field Rleft after a partial energetic reset. Their power spectrum,

$$P_R(k) \propto k^{n_s-1},$$

naturally yields a nearly scale-invariant spectrum with a slight red tilt, consistent with Planck observations.

The acoustic peaks of the CMB are generated by tension-driven oscillations of δR, governed by the wave equation

$$\frac{d^2}{dt^2}\delta R_k + \Gamma_R \frac{d}{dt}\delta R_k + c_R^2 k^2 \delta R_k = 0,$$

which reproduces the observed harmonic structure, odd–even peak asymmetry, and damping tail. Recombination corresponds to a rapid collapse of v, freezing perturbations and imprinting the temperature anisotropies.

A central result of this work is that R-gradients act as effective gravitational sources:

$$\nabla^2\Phi = A_m\,\delta m + A_R\,\delta R,$$

with δRdominating on cosmological scales. This mechanism reproduces the gravitational effects typically attributed to dark matter, including the depth of pre-recombination potential wells, the growth of large-scale structure, and gravitational lensing. TEEM therefore provides a dark-matter-free explanation for these phenomena.

The framework also yields several observational predictions that distinguish it from ΛCDM. These include subtle deviations in the high-ℓ damping tail of the CMB, additional E/B-mode polarization components arising from tension-field dynamics, modified early ISW signatures, earlier and more spatially varied 21-cm absorption features, and characteristic clustering patterns of early massive galaxies. Future high-precision CMB and 21-cm experiments will be able to test these predictions.

Finally, we outlined a numerical framework for implementing TEEM-CMB in a Boltzmann solver, including background evolution, perturbation dynamics, photon propagation, and recombination modeling. This provides a clear path toward quantitative comparison with observational data.

In summary, TEEM-CMB offers a conceptually simple yet dynamically rich alternative to ΛCDM. By grounding cosmology in energetic rather than geometric principles, it reproduces the major observational successes of the standard model while providing new insights into the early universe and the nature of gravitational structure. Further numerical development and observational testing will determine the full viability of this framework as a successor to ΛCDM.

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